

## Mathematics Formula Sheet & Explanation

The 2014 GED® Mathematical Reasoning test contains a formula sheet, which displays formulas relating to geometric measurement and certain algebra concepts. Formulas are provided to test-takers so that they may focus on *application*, rather than the *memorization*, of formulas.

### Area of a:

|               |                               |
|---------------|-------------------------------|
| square        | $A = s^2$                     |
| rectangle     | $A = lw$                      |
| parallelogram | $A = bh$                      |
| triangle      | $A = \frac{1}{2}bh$           |
| trapezoid     | $A = \frac{1}{2}h(b_1 + b_2)$ |
| circle        | $A = \pi r^2$                 |

### Perimeter of a:

|                           |                                                  |
|---------------------------|--------------------------------------------------|
| square                    | $P = 4s$                                         |
| rectangle                 | $P = 2l + 2w$                                    |
| triangle                  | $P = s_1 + s_2 + s_3$                            |
| Circumference of a circle | $C = 2\pi r$ OR $C = \pi d$ ; $\pi \approx 3.14$ |

### Surface area and volume of a:

|                   |                           |                            |
|-------------------|---------------------------|----------------------------|
| rectangular prism | $SA = 2lw + 2lh + 2wh$    | $V = lwh$                  |
| right prism       | $SA = ph + 2B$            | $V = Bh$                   |
| cylinder          | $SA = 2\pi rh + 2\pi r^2$ | $V = \pi r^2 h$            |
| pyramid           | $SA = \frac{1}{2}ps + B$  | $V = \frac{1}{3}Bh$        |
| cone              | $SA = \pi rs + \pi r^2$   | $V = \frac{1}{3}\pi r^2 h$ |
| sphere            | $SA = 4\pi r^2$           | $V = \frac{4}{3}\pi r^3$   |

( $p$  = perimeter of base with area  $B$ ;  $\pi \approx 3.14$ )

### Data

|        |                                                                                                                                                                     |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mean   | mean is equal to the total of the values of a data set, divided by the number of elements in the data set                                                           |
| median | median is the middle value in an odd number of ordered values of a data set, or the mean of the two middle values in an even number of ordered values in a data set |

### Algebra

|                                                |                                                                         |
|------------------------------------------------|-------------------------------------------------------------------------|
| slope of a line                                | $m = \frac{y_2 - y_1}{x_2 - x_1}$                                       |
| slope-intercept form of the equation of a line | $y = mx + b$                                                            |
| point-slope form of the equation of a line     | $y - y_1 = m(x - x_1)$                                                  |
| standard form of a quadratic equation          | $y = ax^2 + bx + c$                                                     |
| quadratic formula                              | $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$                                |
| Pythagorean theorem                            | $a^2 + b^2 = c^2$                                                       |
| simple interest                                | $I = Prt$<br>( $I$ = interest, $P$ = principal, $r$ = rate, $t$ = time) |
| distance formula                               | $d = rt$                                                                |
| total cost                                     | total cost = (number of units) $\times$ (price per unit)                |